Purdue University **2026-2027 University Catalog**

Chemical Engineering, BSCHE

Location(s): West Lafayette

About the Program

The BS program in Chemical Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Chemical, Biochemical, Biomolecular and Similarly Named Engineering Programs. Chemical Engineering remains a premier source of well-educated, well-prepared chemical engineers, educating students using innovative technologies and fostering an environment that inspires leading-edge research.

Chemical engineers work in a wide range of industries with worldwide impact. Applications include energy; pharmaceuticals and biological materials; the nutritional value of food; environmental protection and restoration; materials for computing, sensing, and communications; personal care, home care, and home health products; and system and data management.

Chemical engineers rely on their knowledge of mathematics and science - particularly chemistry - to overcome technical problems in industry and society. While the chemist studies basic chemical reactions, the chemical engineer applies the results of chemical research and transforms laboratory processes into efficient, full-scale processes or facilities. With their strong problem-solving skills and fundamental background in mathematics, physics, chemistry, and biology, chemical engineers can seize opportunities to translate industrial problems into competitive advantages. Currently, chemical engineers demand among the highest salaries for college graduates with a bachelor's degree.

Research here is currently being conducted with polymers and materials, nanoscale science and engineering, fluid mechanics, catalyst design and engineering, sensors, biotechnology, and many others.

Davidson School of Chemical Engineering

Chemical Engineering Major Change (CODO) Requirements

Degree Requirements

130 Credits Required

Chemical Engineering Major Requirements (46 credits)

Chemical Engineering Required Major Courses (43 credits)

A grade of C- or better is required in all Required Major Courses and Chemical Engineering Selective, unless a specific grade is listed for the course.

- CHE 20000 - Chemical Engineering Seminar **Credit Hours: 1.00**
- CHE 20500 - Chemical Engineering Calculations **Credit Hours: 4.00**
- CHE 21100 - Introductory Chemical Engineering Thermodynamics **Credit Hours: 4.00**
- CHE 30000 - Chemical Engineering Seminar **Credit Hours: 1.00**
- CHE 30600 - Design Of Staged Separation Processes **Credit Hours: 3.00**
- CHE 32000 - Statistical Modeling And Quality Enhancement **Credit Hours: 3.00**
- CHE 34800 - Chemical Reaction Engineering **Credit Hours: 4.00**
- CHE 37700 - Momentum Transfer **Credit Hours: 4.00**
- CHE 37800 - Heat And Mass Transfer **Credit Hours: 4.00**
- CHE 40000 - Chemical Engineering Seminar **Credit Hours: 1.00**
- CHE 42000 - Process Safety Management And Analysis **Credit Hours: 3.00** ♦
- CHE 43500 - Chemical Engineering Laboratory **Credit Hours: 4.00**
- CHE 45000 - Design And Analysis Of Processing Systems **Credit Hours: 4.00**
- CHE 45600 - Process Dynamics And Control **Credit Hours: 3.00** ♦

Chemical Engineering Selective (3 credits)

- Chemical Engineering Selective - Credit Hours: 3.00 (see Supplemental Information)

Other Departmental/Program Course Requirements (87-97 credits)

Other Course Requirements (36-40 credits)

Intro to Engineering I & II - Credit Hours: 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

OR

- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and

- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

  OR
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - **2.00 Credit Hours**

  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

  OR
- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

  AND
- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- CHM 11610 - General Chemistry II **Credit Hours: 3.00**
- CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00**
- CHM 26100 - Organic Chemistry I **Credit Hours: 3.00** ♦
- CHM 26300 - Organic Chemistry Laboratory I **Credit Hours: 1.00** ♦
- CHM 26200 - Organic Chemistry II **Credit Hours: 3.00** ♦
- CHM 26400 - Organic Chemistry Laboratory II **Credit Hours: 1.00** ♦
- CHM 37000 - Topics In Physical Chemistry **Credit Hours: 3.00** ♦
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** ♦
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

Biology Selective - Credit Hours: 3.00

- BCHM 30700 - Biochemistry **Credit Hours: 3.00**
- BCHM 56600 - General Biochemistry **Credit Hours: 4.00**
- CHM 33900 - Biochemistry: A Molecular Approach **Credit Hours: 3.00**

- CHM 43300 - Biochemistry **Credit Hours: 3.00**

Physics I - Credit Hours: 4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00** **Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00** **Credit Hours**

- Engineering Selective I - Credit Hours: 3.00
- Engineering Selective II - Credit Hours: 3.00
- Math Selective I - Credit Hours: 3.00
- Math Selective II - Credit Hours: 3.00
- Technical Selective - Credit Hours: 3.00

Engineering General Education Requirements (24-25 credits)

For a list of courses that satisfy this requirement, see Supplemental Information

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements

- At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)
- At least 12 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.
- A course is not eligible for use towards the 24 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.

Supplemental Information

- Chemical Engineering Supplemental Information

Optional Concentrations

- Biological Engineering Concentration in Chemical Engineering
- Data Science Concentration in Chemical Engineering
- Energy and The Environment Concentration in Chemical Engineering
- Materials and Polymers Concentration in Chemical Engineering
- Pharmaceutical Engineering Concentration for Chemical Engineering
- Research in Chemical Engineering Concentration for Chemical Engineering
- Semiconductors Concentration for Chemical Engineering

GPA Requirements

- A GPA of 2.0 is required for all Baccalaureate degrees.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.

*If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

Grade Requirements

- Students must earn a “C” or higher in CHE 20500 to enroll in any further CHE courses.
- A grade of C- or better is required in all Required Major Courses, Chemical Engineering Selective, and Math courses, unless a specific grade is listed for the course.

Pass/No Pass Policy

- CHE students may take the Engineering General Education required courses for a letter grade or pass/ no pass option. All other courses may not be taken as P/NP.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00

- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- CHM 11610 - General Chemistry II **Credit Hours: 3.00**
- CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

17-21 Credits

Fall 2nd Year

- CHE 20000 - Chemical Engineering Seminar **Credit Hours: 1.00**
- CHE 20500 - Chemical Engineering Calculations **Credit Hours: 4.00**
- CHM 26100 - Organic Chemistry I **Credit Hours: 3.00** ♦
- CHM 26300 - Organic Chemistry Laboratory I **Credit Hours: 1.00** ♦
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** ♦

16 Credits

Spring 2nd Year

- CHE 21100 - Introductory Chemical Engineering Thermodynamics **Credit Hours: 4.00**
- CHE 32000 - Statistical Modeling And Quality Enhancement **Credit Hours: 3.00**
- CHM 26200 - Organic Chemistry II **Credit Hours: 3.00** ♦
- CHM 26400 - Organic Chemistry Laboratory II **Credit Hours: 1.00** ♦
- Math Selective I - Credit Hours: 3.00
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

17 Credits

Fall 3rd Year

- CHE 30600 - Design Of Staged Separation Processes **Credit Hours: 3.00**
- CHE 37700 - Momentum Transfer **Credit Hours: 4.00**
- CHM 37000 - Topics In Physical Chemistry **Credit Hours: 3.00** ♦
- Math Selective II - Credit Hours: 3.00
- Biology Selective - Credit Hours: 3.00

16 Credits

Spring 3rd Year

- CHE 30000 - Chemical Engineering Seminar **Credit Hours: 1.00**
- CHE 37800 - Heat And Mass Transfer **Credit Hours: 4.00**
- CHE 34800 - Chemical Reaction Engineering **Credit Hours: 4.00**
- Engineering Selective I - Credit Hours: 3.00
- Technical Selective - Credit Hours: 3.00
- General Education II (UCC: BSS) - Credit Hours: 3.00

18 Credits

Fall 4th Year

- CHE 40000 - Chemical Engineering Seminar **Credit Hours: 1.00**

- CHE 42000 - Process Safety Management And Analysis **Credit Hours: 3.00**
- CHE 43500 - Chemical Engineering Laboratory **Credit Hours: 4.00**
- CHE 45600 - Process Dynamics And Control **Credit Hours: 3.00**
- General Education III – Credit Hours: 3.00
- General Education IV – Credit Hours: 3.00

17 Credits

Spring 4th Year

- CHE 45000 - Design And Analysis Of Processing Systems **Credit Hours: 4.00**
- Chemical Engineering Selective - Credit Hours: 3.00
- Engineering Selective II - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- General Education VI - Credit Hours: 3.00

16 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.